Unit II :: Mathematical Typesetting and Technical Elements in LaTeX

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Syllabus

Unit II: Mathematical Typesetting and Technical Elements in LaTeX

Math mode: inline and display modes; math fonts: mathcal, mathbb, mathds, mathfrak; superscripts and subscripts, pre-defined functions, fractions, roots, Greek letters; integrals, summations and products; mathematical grids: multi-line formulae, gathering system of equations, aligning system of equations, matrices; LaTeX packages for Physics: siunitx, physics; floats in LaTeX: table and figure environments; typesetting tables using tabular; formatting figures using includegraphics.

1 Overview of math mode in LaTeX

In LaTeX, the typesetting of mathematical expressions requires entering a dedicated operating environment known as *math mode*. In math mode, LaTeX does the following

- Treats alphabet symbols as mathematical variables.
- Typesets letters in italics by default.
- Ignores standard whitespace characters.
- Adjusts spacing between symbols dynamically based on their semantic classification, such as operators, relations and delimiters.

1.1 Inline math mode

Inline math mode is used to embed short mathematical expressions and variables directly within a sentence without disrupting the paragraph flow. We can typeset a mathematical expression in inline mode using any of the following syntax:

1. Enclosing the expression within `$...$`, for example `$F=ma$`.
2. Enclosing the expression within `\(...\)`, for example `\(p=mv\)`.
3. Placing the expression within the `math` environment, for example
```
\begin{math}
    U=mgh
\end{math}
```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $F=
ma$, linear momentum \(p=mv\), and gravitational potential energy
\begin{math}
  U=mgh
\end{math}.
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion $F = ma$, linear momentum $p = mv$, and gravitational potential energy $U = mgh$.

1.2 Display math mode

Display math mode is used for prominent, complex or multi-line equations that require dedicated vertical separation, centered alignment and full symbol scaling.

For display mode, we have two types of syntax:

1. *Un-numbered display syntax*:

   (a) Enclosing the expression within `$$...$$`, for example `$$F=ma$$`.

   (b) Enclosing the expression within `\[...\]`, for example `\[p=mv\]`.

   (c) Placing the expression within the `displaymath` environment, for example
   ```
   \begin{displaymath}
       U=mgh
   \end{displaymath}
   ```

2. *Numbered display syntax*: Placing the expression within the `equation` environment, in which case the equation will also be given a number. For example
   ```
   \begin{equation}
       T=\frac{1}{2}mv^2
   \end{equation}
   ```
   Note: The numbering of the equation can be suppressed by using the `equation*` environment, with an asterisk (*) appended. For example
   ```
   \begin{equation*}
       E=mgh+\frac{1}{2}mv^2
   \end{equation*}
   ```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $$F=
ma$$ linear momentum \[p=mv\] gravitational potential energy
\begin{displaymath}
  U=mgh
\end{displaymath}
and kinetic energy
\begin{equation}
  T=\frac{1}{2}mv^2
\end{equation}
with the mechanical energy
\begin{equation*}
  E=mgh+\frac{1}{2}mv^2
\end{equation*}
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion

$$F = ma$$

linear momentum

$$p = mv$$

gravitational potential energy

$$U = mgh$$

and kinetic energy

$$T = \frac{1}{2}mv^2 \tag{1}$$

with the mechanical energy

$$E = mgh + \frac{1}{2}mv^2$$

2 Mathematical typography in LaTeX

2.1 Superscripts and subscripts

Superscripts and subscripts are commonly used to represent exponents (such as 3^2, x^{-3}) and indexed quantities (such as m_1, K_0) respectively. In LaTeX, superscripts are typeset using the *caret* symbol (`^`) while subscripts are typeset using the *underscore* symbol (`_`).

If a superscript or subscript contains more than a single character, the entire exponent or index must be enclosed in braces `{ }`.

For instance, the following code contains expressions with exponents and indices

```
$$ax^2+bx+c=0$$

$$f(x)=e^{-ax^2}$$

$$m_1=5, m_{12}=36.5$$
```

which produces the output

$$a x^2 + b x + c = 0$$

$$f(x) = e^{-a x^2}$$

$$m_1 = 5, m_{12} = 36.5$$

2.2 Fractions and roots

In LaTeX, fractions are typeset using the `\frac` macro. The general syntax is

```
\frac{numerator}{denominator}
```

For fractions contained inline, where vertical space is limited, standard slash notation `a/b` or the `\tfrac{num}{den}` macro can be used.

For instance, the following code contains examples of typesetting fractions

```
The kinetic energy of a particle of mass $m$ and moving with speed $v$ is $\
frac{1}{2}mv^2$.

Using the law of equipartition of energy, the average energy of a monoatomic
molecule is obtained to be $\tfrac{3}{2}k_BT$.
```

which produces the output

The kinetic energy of a particle of mass m and moving with speed v is $\frac{1}{2}mv^2$.
Using the law of equipartition of energy, the average energy of a monoatomic molecule is obtained to be $\frac{3}{2}k_BT$.

LaTeX allows us to typeset the square root using the `\sqrt` macro which has the general syntax as

```
\sqrt{expression}
```

The `\sqrt` macro can be extended to typeset the nth root using an optional argument. The general syntax for this is

```
\sqrt[n]{expression}
```

For instance, the following code contains examples of typesetting roots

```
The root mean squared speed of molecules is $\sqrt{\dfrac{3k_BT}{m}}$, as per
Maxwell-Boltzmann distribution.

The roots of a quadratic equation are
\begin{equation}
  x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}
\end{equation}
```

which produces the output

The root mean squared speed of molecules is $\sqrt{\dfrac{3k_BT}{m}}$, as per Maxwell-Boltzmann distribution.
The roots of a quadratic equation are

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \tag{2}$$

2.3 Pre-defined functions and operators

Standard mathematical functions (such as trigonometric, logarithmic and limit operators) must not be typed as plain text in math mode; otherwise, LaTeX would render them in variable italics. For example, `sin(x)` typesets as a product of s, i and n. Hence to render this correctly in upright roman typeface with appropriate surrounding whitespace, LaTeX provides pre-defined operator macros for a bunch of functions as given below.

- *Trigonometric and hyperbolic functions*: `\sin`, `\cos`, `\tan`, `\arcsin`, `\arccos`, `\arctan`, `\sinh`, `\cosh`, `\tanh`
- *Logarithmic and exponential functions*: `\exp`, `\log`, `\ln`
- *Limits and optimization*: `\lim`, `\min`, `\max`, `\sup`, `\inf`
- *Linear algebra*: `\det`, `\ker`, `\deg`

2.4 Summation, product and integral operators

In LaTeX, the summation, product and integral operators are typeset using the \sum, \prod and \int macros. These are variable-sized operators and get rendered in different sizes, depending on whether they are in the inline or display modes.

The following code typesets these in the inline mode:

```
$\sum_{i=1}^{N}(2i-1)$, $\prod_{k=1}^{N}\frac{1}{2^k}$ and $\int_{0}^{5}e^{-x}
dx$
```

which gives the following output:

$\sum_{i=1}^{N}(2i-1)$, $\prod_{k=1}^{N}\frac{1}{2^k}$ and $\int_{0}^{5}e^{-x}dx$

In display math mode, variable-sized operators render larger than in inline mode, and their limits adjust automatically. The following code typesets the above expressions in the display mode:

```
$$\sum_{i=1}^{N}(2i-1)$$ $$\prod_{k=1}^{N}\frac{1}{2^k}$$ $$\int_{0}^{5}e^{-x}
dx$$
```

which gives the following output:

$$\sum_{i=1}^{N}(2i-1)$$

$$\prod_{k=1}^{N}\frac{1}{2^k}$$

$$\int_{0}^{5}e^{-x}dx$$

2.5 Greek letters

In physics, letters from the Greek alphabet are widely used to represent various physical quantities. In LaTeX, lowercase Greek letters are typeset by prefixing the letter's English name with a backslash, as given in the following table:

Letter	Macro	Letter	Macro	Letter	Macro	Letter	Macro
α	\alpha	η	\eta	ν	\nu	υ	\upsilon
β	\beta	θ	\theta	ξ	\xi	ϕ	\phi
γ	\gamma	ι	\iota	π	\pi	χ	\chi
δ	\delta	κ	\kappa	ρ	\rho	ψ	\psi
ϵ	\epsilon	λ	\lambda	σ	\sigma	ω	\omega
ζ	\zeta	μ	\mu	τ	\tau		

For the uppercase Greek letters that do not resemble any of the letters from the English alphabet, the typesetting is done in LaTeX by capitalizing the first letter of the macro for the same lowercase

letter. For instance, when the macro for the lowercase lambda λ, which is `\lambda`, is capitalized as `\Lambda`, we obtain the uppercase lambda Λ.

The following table provides the macros for the uppercase Greek letters.

Letter	Macro	Letter	Macro	Letter	Macro	Letter	Macro
Γ	`\Gamma`	Λ	`\Lambda`	Σ	`\Sigma`	Ψ	`\Psi`
Δ	`\Delta`	Ξ	`\Xi`	Υ	`\Upsilon`	Ω	`\Omega`
Θ	`\Theta`	Π	`\Pi`	Φ	`\Phi`		

2.6 Mathematical fonts

To represent specific mathematical structures (such as sets, vector spaces, or Lie algebras), LaTeX provides the following specialized math font commands.

Math font class	Package	Macro	Example	Result
Calligraphic	`amsmath`	`\mathcal`	`\mathcal{A,B,C}`	$\mathcal{A}, \mathcal{B}, \mathcal{C}$
Blackboard bold	`amssymb`	`\mathbb`	`\mathbb{R,N,Z}`	$\mathbb{R}, \mathbb{N}, \mathbb{Z}$
Double stroke	`dsfont`	`\mathds`	`\mathds{P,Q,R}`	$\mathbb{P}, \mathbb{Q}, \mathbb{R}$
Fraktur/Gothic	`eufrak`	`\mathfrak`	`\mathfrak{g,H,K}`	$\mathfrak{g}, \mathfrak{H}, \mathfrak{K}$

3 Alignment of mathematical expressions

The amsmath package provides specialized environments that allow us to structure sophisticated mathematical expressions by splitting them across multiple lines and aligning multiple equations in grids.

To use the amsmath package, we include the following line in the preamble:

```
\usepackage{amsmath}
```

3.1 Splitting an equation into multiple lines

Suppose we have a rather long equation whose length exceed the page width, such as that using the code below:

```
\begin{equation}
  (a+b)^{12}=a^{12}+12a^{11}b+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5+924a^6b
^6+792a^5b^7+495a^4b^8+220a^3b^9+66a^2b^{10}+12ab^{11}+b^{12}
\end{equation}
```

which yields the following output that exceeds the margins of the page.

$$(a+b)^{12} = a^{12}+12a^{11}b+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9+ \tag{1}$$

In such scenario, the multline environment can be used, where the equation can be split using double back-slash (\\). In this environment, the first line is left-aligned, the final line is right-aligned and intermediate lines are centered. The above long equation can be re-written as shown below:

```
\begin{multline}
  (a+b)^{12}=a^{12}+12a^{11}b \\
  +66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  +924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
  +66a^2b^{10}+12ab^{11}+b^{12}
\end{multline}
```

which gives the following output

$$\begin{aligned}(a+b)^{12} = a^{12} + 12a^{11}b \\ &+ 66a^{10}b^2 + 220a^9b^3 + 495a^8b^4 + 792a^7b^5 \\ &+ 924a^6b^6 + 792a^5b^7 + 495a^4b^8 + 220a^3b^9 \\ &+ 66a^2b^{10} + 12ab^{11} + b^{12}\end{aligned} \tag{1}$$

However if we want all the lines of the equation to be centre aligned instead, then we can use the `gathered` environment wrapped within an outer `equation` environment, as shown below:

```
\begin{equation}
  \begin{gathered}
    (a+b)^{12}=a^{12}+12a^{11}b \\
    +66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
    +924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
    +66a^2b^{10}+12ab^{11}+b^{12}
  \end{gathered}
\end{equation}
```

which yields the following output:

$$\begin{gathered}(a+b)^{12}=a^{12}+12a^{11}b \\ +66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\ +924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\ +66a^2b^{10}+12ab^{11}+b^{12}\end{gathered} \tag{1}$$

3.2 Aligning multiple equations

The most effective way to collect and align multiple equations is by using the `align` environment. In this environment, the expressions in different lines are aligned at specified horizontal anchor points marked by ampersands (`&`). The lines are broken into rows using double back-slashes (`\\`).

For example, the long equation from the preceding sub-section can be split and aligned using the `align` environment as shown below:

```
\begin{align}
  (a+b)^{12}&=a^{12}+12a^{11}b \\
  &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
  &+66a^2b^{10}+12ab^{11}+b^{12}
\end{align}
```

which yields the following output:

$$\begin{aligned}(a+b)^{12}&=a^{12}+12a^{11}b && (1)\\ &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 && (2)\\ &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 && (3)\\ &+66a^2b^{10}+12ab^{11}+b^{12} && (4)\end{aligned}$$

As it can be observed, the `align` environment numbers each line with unique equation numbers. Such numbering can be suppressed for all the lines by using the `align*` instead, as shown below:

```
\begin{align*}
  (a+b)^{12}&=a^{12}+12a^{11}b \\
  &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
  &+66a^2b^{10}+12ab^{11}+b^{12}
\end{align*}
```

which gives the output:

$$\begin{aligned}(a+b)^{12} &= a^{12}+12a^{11}b \\ &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\ &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\ &+66a^2b^{10}+12ab^{11}+b^{12}\end{aligned}$$

In case we wish to suppress the numbering only on specific lines, then in the `align` environment we can use the `\nonumber` macro just before the `\\` for that line. For example, consider the following code:

```
\begin{align}
  (a+b)^{12}&=a^{12}+12a^{11}b \nonumber\\
  &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \nonumber\\
  &+66a^2b^{10}+12ab^{11}+b^{12} \nonumber
\end{align}
```

which gives the following output:

$$\begin{aligned}(a+b)^{12} &= a^{12}+12a^{11}b \\ &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\ &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\ &+66a^2b^{10}+12ab^{11}+b^{12}\end{aligned} \tag{1}$$

Note: On a personal note, I prefer to use the `\align` environment as a one-stop solution for typesetting virtually all of the mathematical expressions. This is because of the versatility which it affords me in terms of aligning expressions and toggling equation numbering on and off.

Consequently, I have found the `\align` environment to be helpful in the display math mode, particularly for the following scenarios:

1. Typesetting a system of equations. For instance, consider the following code:

```
\begin{align*}
  x-3y+4z&=9 \\
  2x+5y+z&=7 \\
  -4x-y+z&=3
\end{align*}
```

   which yields the following system of linear equations:

$$\begin{aligned} x-3y+4z&=9 \\ 2x+5y+z&=7 \\ -4x-y+z&=3 \end{aligned}$$

2. Writing out the steps in a derivation. For instance, consider the following code:

```
\begin{align*}
  F&\propto\frac{dp}{dt} \\
  &=k\frac{dp}{dt} \\
  &=k\frac{d}{dt}(mv) \\
  &=km\frac{dv}{dt} \\
  \implies F&=kma
\end{align*}
```

   which yields the following simple derivation:

$$\begin{aligned} F &\propto \frac{dp}{dt} \\ &= k\frac{dp}{dt} \\ &= k\frac{d}{dt}(mv) \\ &= km\frac{dv}{dt} \\ \implies F &= kma \end{aligned}$$

4 Typesetting matrices in LaTeX

The `amsmath` package provides commands to typeset matrices with different delimiters. Once the `amsmath` package has been included in the preamble, different types of matrices can be typeset using different environments. Each of these environments have to be wrapped inside an outer math environment, such as `equation` and `align`.

The general syntax is shown below:

```
\begin{matrix environment name}
  {row 1, col 1} & {row 1, col 2} & {...} & {row 1, col n} \\
  {row 2, col 1} & {row 2, col 2} & {...} & {row 2, col n} \\
  {...} & {...} & {...} & {...} \\
  {row m, col 1} & {row m, col 2} & {...} & {row m, col n}
\end{matrix environment name}
```

4.1 Plain matrix

A *plain matrix* is one in which the elements are arranged in rows and columns without any enclosing delimiters. For instance, the following code is used to typeset a plain matrix:

```
\begin{equation*}
  \begin{matrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{matrix}
\end{equation*}
```

which gives the following output:

$$\begin{matrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{matrix}$$

4.2 Parentheses matrix

A *parentheses matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized parentheses `( )` as delimiters. For instance, the following code is used to typeset a parentheses matrix:

```
\begin{equation*}
  \begin{pmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{pmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{pmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{pmatrix}$$

4.3 Brackets matrix

A *brackets matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized brackets `[ ]` as delimiters. For instance, the following code is used to typeset a brackets matrix:

```
\begin{equation*}
  \begin{bmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{bmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{bmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{bmatrix}$$

4.4 Braces matrix

A *braces matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized braces `{ }` as delimiters. For instance, the following code is used to typeset a braces matrix:

```
  \begin{equation*}
    \begin{Bmatrix}
      {1} & {-9} & {5} \\
      {4} & {8} & {-2} \\
      {-6} & {3} & {7}
    \end{Bmatrix}
  \end{equation*}
```

which gives the following output:

$$\begin{Bmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{Bmatrix}$$

4.5 Pipes matrix

A *pipes matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized pipes | | as delimiters. For instance, the following code is used to typeset a pipes matrix:

```
\begin{equation*}
  \begin{vmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{vmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{vmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{vmatrix}$$

4.6 Double-pipes matrix

A *double pipes matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized double-pipes || || as delimiters. For instance, the following code is used to typeset a double-pipes matrix:

```
\begin{equation*}
  \begin{Vmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{Vmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{Vmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{Vmatrix}$$

5 LaTeX packages for Physics

5.1 The `siunitx` package

The `siunitx` package provides tools for typesetting physical quantities, numbers, and units consistently in LaTeX. Maintained by Joseph Wright, the current version (v3) provides extensive automated formatting, parsing, rounding, and decimal-aligned table columns.

To use the `siunitx` package, we include the following line in the preamble:

```
\usepackage{siunitx}
```

The `siunitx` package enforces the BIPM (International Bureau of Weights and Measures) standards for formatting numbers and units. For physics lab reports and dissertations, it prevents critical formatting errors such as italicizing units, misaligning decimals or using incorrect spacing between values and units.

Macros from the `siunitx` package work in both text and math modes. The package's engine auto-detects the surrounding environment and ensures that the numbers and units are always typeset in upright roman fonts, regardless of where the macros are placed.

5.1.1 Fundamental macros

The fundamental macros provided by `siunitx` handles the spacing as well as the font selection. These are briefly summarised below:

1. *Standalone numbers* can be typeset using the `\num{value}` macro, where `value` is the numerical value of the number.

2. *Standalone units* can be typeset using the `\unit{unit macro}` macro, where `unit macro` could be a combination of macros provided by the package.

3. *Full quantities*, consisting of numbers and units can be typeset using the `\qty{value}{unit macro}`, which takes in two arguments, which are the same as for the standalone number and unit.

5.1.2 Scientific notation and uncertainties

Manually handling scientific notation and error values in LaTeX requires tedious formatting. The `siunitx` package can parse raw inputs and format them accordingly.

Scientific notation: Numbers are formatted in the correct scientific notation by passing the value argument in base-10 exponent format. For instance, `\num{6.67e-11}` is typeset in the scientific notation as 6.67×10^{-11}.

Uncertainties: Errors or uncertainties in quantities can be appended to the value argument using the `\pm` macro and setting the optional argument `separate-uncertainty=true`. For instance, `\num[separate-uncertainty=true]{9.81\pm0.03}` is typeset with the value and uncertainty as 9.81 ± 0.03.

To avoid writing `[separate-uncertainty=true]` repeatedly, we can set it globally for the document by including the following line in the preamble:

```
\sisetup{separate-uncertainty=true}
```

After this, simply typing `\num{9.81\pm0.03}` ensures correct typesetting to 9.81 ± 0.03.

5.2 The `physics` package

The `physics` package was a popular LaTeX package created by Sergio C. de la Barrera that simplified and standardized the typesetting of physics, vector calculus and linear algebra equations.

To use the `physics` package, we include the following line in the preamble:

```
\usepackage{physics}
```

It must be noted that the macros from the `physics` package work in the math mode alone.

The package is designed to provide shorthand macros for tedious physics notation. Its core functionality covers four main areas:

1. **Vectors:**
   - Different representations of vectors can be typeset using the `$\va{v}$` and `$\vb{A}$` macros which are rendered as $\vec{\mathbf{v}}$ and $\mathbf{A}$ respectively.
   - Unit vectors are typeset using the `$\vu{n}$` macro to give $\hat{\mathbf{n}}$. Another variation is the `$\vu*{n}$` macro which gives the italicized $\hat{\boldsymbol{n}}$.
2. **Differentials and derivatives:**
   - For differentials, the `$\dd{x}$` macro typesets as $\mathrm{d}x$.
   - Total derivatives are typeset using the `$\dv{f}{x}$` macro to give $\frac{\mathrm{d}f}{\mathrm{d}x}$.
   - Partial derivatives are typeset using the `$\pdv{f}{x}$` macro to give $\frac{\partial f}{\partial x}$.
   - Higher order derivatives are typeset by passing the order of the derivatives as an optional argument. For instance, `$\dv[3]{f}{x}$` and `$\pdv[2]{f}{x}{x}$` typeset as $\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}$ and $\frac{\partial^2 f}{\partial x \partial y}$ respectively.
3. **Vector calculus:** There are shorthands for the del (∇) operator in the form of the `$\grad{f}$`, `$\div{\vb{v}}$` and `$\curl{\vb{v}}$` macros which typeset as $\boldsymbol{\nabla} f$, $\boldsymbol{\nabla} \cdot \mathbf{v}$ and $\boldsymbol{\nabla} \times \mathbf{v}$ respectively.
4. **Variable-sized delimiters:** Using the `\qty` macro along with different delimiters, causes the delimiters to be automatically sized so as to match the contents within.

   For instance, with brackets using `$\qty[F-ma]$` and `$\qty[F-\dv{v}{t}]$` gives $[F - ma]$ and $\left[F - \frac{\mathrm{d}v}{\mathrm{d}t}\right]$ respectively. Likewise, for other delimiters.

6 Floating environments in LaTeX

Certain document elements, such as large tables and graphic diagrams, cannot be broken across pages. If placed directly in the text stream, they could leave awkward blank gaps at the bottom of pages.

To resolve this, LaTeX places them inside floating containers or *floats*, allowing the compiler to move them to optimal page positions without breaking text flow. In other words, LaTeX allows them to "float" across the document to optimum positions.

There are two floating environments which are commonly used in LaTeX:

1. The `table` floating environment which is used to construct tables within the document.
2. The `figure` floating environment which is used to incorporate figures within the document.

Position specifiers: To guide LaTeX as to the order of preference where we want the float to be laid out in the document, we can use any or a combination of the following optional placement parameters:

- `h` for *here* which tries to place it approximately at the location in the source code.
- `t` for *top* which tries to place it at the top of a page.
- `b` for *bottom* which tries to place it at the bottom of a page.
- `p` for *page* which tries to place it on a dedicated float-only page.
- `!` for *override* which tries to relaxe internal LaTeX placement restrictions for float count and area.

For instance, when we combine the placement specifiers as `htbp`, LaTeX first tries to place the float at the location in the *source code* (because of the `h`). If that is not possible, then it tries to place the float at the *top* of the current or next page (because of the `t`). Again, if that is not possible, it tries to place the float at the *bottom* of the current or next page (because of the `b`). Finally, if all of the above fail, then it places the float on a *page dedicated to floats* only (because of the `p`), in the order in which they appear in the document.

Captions, labels and referencing: Given below are the macros which are available for including captions for the float, adding labels and referencing them elsewhere.

- `\caption{text}` generates a numbered caption for the table or figure.
- `\label{key}` assigns a symbolic key value to the element so that it can be referenced elsewhere. The `\label` macro must always be placed below the `\caption` macro.
- `\ref{key}` and `\pageref{key}` retrieve the float number and the page number where it currently appears in the document.

6.1 Typesetting tables

LaTeX allows us to typeset tables using the `tabular` environment which is wrapped within a `table` floating environment.

The general syntax for using the `tabular` environment is structured as follows:

```
\begin{table}
  \caption{table caption}
  \label{key}
  \begin{tabular}[position]{column specifiers}
    {cell 11} & {cell 12} & {...} & {cell 1n} \\
    {cell 21} & {cell 22} & {...} & {cell 2n} \\
    {...} & {...} & {...} & {...} \\
    {cell m1} & {cell m2} & {...} & {cell mn} \\
  \end{tabular}
\end{table}
```

6.1.1 Column specifiers

This is a mandatory sequence of formatting characters defining the number, alignment and border layout of columns.

- `l` for *left*-aligned column.
- `c` for *centre*-aligned column.
- `r` for *right*-aligned column.
- | to insert a vertical gridline between adjacent columns.
- `p{width}` for a fixed-width *paragraph* column that automatically wraps multi-line text.

6.1.2 Table body

The body of the table, which contains the rows and columns, has a syntax which requires the following elements:

- The column separator (&) separates individual cells within a row.
- The row terminator (\\) marks the end of a row. An optional length argument (e.g. `\\[5pt]`) can be added to insert extra vertical space after a row.
- The full horizontal gridline (`\hline`) draws a horizontal rule across the full width of the table.
- The partial horizontal gridline (`\cline{i-j}`) draws a horizontal rule from column `i` to column `j` of the table.

6.1.3 Vertical position

This is an optional argument which controls how the entire table aligns vertically relative to the baseline of the surrounding narrative text.

- `c`, which is the default value, aligns the vertical *centre* of the table with the text baseline.
- `t` aligns the baseline of the *top* row with the surrounding text baseline.
- `b` aligns the baseline of the *bottom* row with the surrounding text baseline.

For instance, the following code typesets a table:

```
\begin{table}[h]
  \caption{Table of voltage and current observations}
  \label{tab:ohms-law}
  \begin{tabular}[t]{|c|c|c|c|}
    \hline
    \textbf{S. No.} & \textbf{Voltage (V)} & \textbf{Current (mA)} & \textbf{
Resistance ($\Omega$)} \\ \hline
    {1} & {0.5} & {3.5} & {142} \\ \hline
    {2} & {1.0} & {6.7} & {149} \\ \hline
    {3} & {1.5} & {9.9} & {151} \\ \hline
    {4} & {2.0} & {13.1} & {153} \\ \hline
    {5} & {2.5} & {17.2} & {145} \\ \hline
  \end{tabular}
\end{table}
```

which yields the following output:

Table 1: Table of voltage and current observations

S. No.	**Voltage (V)**	**Current (mA)**	**Resistance (Ω)**
1	0.5	3.5	142
2	1.0	6.7	149
3	1.5	9.9	151
4	2.0	13.1	153
5	2.5	17.2	145

6.2 Formatting figures

To incorporate external vector graphics (e.g. PDF, EPS) or raster images (e.g. PNG, JPG), we can use the `\includegraphics` macro which is provided by the `graphicx` package.

To use the `graphicx` package, we include the following line in the preamble:

```
\usepackage{graphicx}
```

The general syntax for using the `\includegraphics` macro is structured as follows:

```
\begin{figure}
  \centering
  \includegraphics[key=value-list]{filename}
  \caption{table caption}
  \label{key}
\end{figure}
```

The various arguments used are as follows:

- The mandatory argument `filename` contains the relative or absolute path of the graphic file. The file extension (e.g. `.png`, `.jpg`, `.pdf`, `.eps`) can be omitted, as LaTeX automatically searches for supported formats.
- The optional arguments `key=value-list` is a comma-separated list of properties to resize, scale, rotate or trim the graphic. Some of the commonly used parameters are
  - `width` which resizes the image to a specific width (e.g. `width=4.5cm` or `width=0.8\textwidth`).
  - `height` which resizes the image to a specific height (e.g. `height=3.6in` or `width=0.5\textheight`).
  - `keepaspectratio` which prevents image distortion when both `width` and `height` are specified simultaneously.
  - `scale` which scales both width and height proportionally by a factor (e.g. `scale=0.6` scales the image to 60% of its size).
  - `angle` which rotates the image counter-clockwise by a specified angle in degrees (e.g. `angle=45`).
  - `trim` which crops margins off the image as `trim=left bottom right top` (e.g. `trim =1cm 2cm 1cm 2cm`).

For instance, the following code incorporates a figure in the file named `schrodingers-cat.png` which is contained in a folder titled `figures` that lies in the same location as the LaTeX source code:

```
\begin{figure}[h!]
  \centering
  \includegraphics[width=0.4\textwidth, angle=30]{figures/schrodingers-cat}
  \caption{Schrodinger's cat}
  \label{fig:schrodinger-cat}
\end{figure}
```

which yields the following output:



Figure 1: Schrodinger's cat